

2. Common Elements Between Two Arrays and find its complexity

Aim

To write a Python program to find the **common elements between two arrays** and analyze its time and space complexity.

Algorithm

1. Start.
2. Read two arrays.
3. Compare each element of the first array with elements of the second array.
4. If an element is present in both arrays, add it to the result.
5. Print the common elements.
6. Stop.

Source Code

```
return result def common_elements(arr1, arr2):  
    result = []  
    for x in arr1:  
        if x in arr2 and x not in result:  
            result.append(x)  
    arr1 = [1, 2, 3, 4, 5]  
    arr2 = [3, 4, 5, 6, 7]  
    print("Common Elements:", common_elements(arr1, arr2))
```

Output

```
main.py Visualize Run ▶ Output
1 def common_elements(arr1, arr2):
2     result = []
3
4     for x in arr1:
5         if x in arr2 and x not in result:
6             result.append(x)
7
8     return result
9
10
11 arr1 = [1, 2, 3, 4, 5]
12 arr2 = [3, 4, 5, 6, 7]
13
14 print("Common Elements:", common_elements(arr1, arr2))
```

Common Elements: [3, 4, 5]

Complexity

Let:

- n = number of elements in the first array
- m = number of elements in the second array

Time Complexity: $O(n \times m)$ in the worst case.

Space Complexity: $O(k)$, where k is the number of common elements.

Result

Thus, the Python program successfully finds the **common elements between two arrays**. The worst-case time complexity is **$O(n \times m)$** .